

Memory Management, Virtual Memory & Storage - Detailed Notes

DETAILED COMPREHENSIVE NOTES

Subject: Operating Systems | Chapter: Chapter 3: Memory Management, Virtual Memory & File Systems | Academic Year: 2026-2027

1. Logical versus Physical Address Space & Paging

Logical addresses are generated by the CPU during execution; Physical addresses are loaded into memory hardware. The Memory Management Unit (MMU) translates logical to physical addresses at runtime.

Paging eliminates external fragmentation by dividing physical memory into fixed-size Frames and logical memory into same-sized Pages. The Page Table translates Page Number (p) to Frame Number (f), maintaining Page Offset (d).

Translation Lookaside Buffer (TLB): A hardware associative cache storing recent page translations to reduce memory access latency.

2. Virtual Memory & Page Replacement Algorithms

Virtual Memory allows execution of processes that are not completely in physical memory, providing larger addressable memory than physical RAM through Demand Paging.

When a page is accessed that is not in memory (valid-invalid bit is invalid), the MMU triggers a Page Fault interrupt.

Page Replacement Algorithms: FIFO (suffers from Belady's Anomaly where more frames increase page faults), Optimal (replaces page not used for longest future duration, theoretical benchmark), and LRU (Least Recently Used, practical approximation using timestamps or stack).

3. File System Architecture & Disk Scheduling

File systems structure data hierarchically. In Unix, an Inode stores file metadata (permissions, owner, size, timestamps) and direct/indirect block pointers.

Disk scheduling algorithms optimize hard drive arm movement: FCFS, SSTF (Shortest Seek Time First, can cause starvation), SCAN (Elevator algorithm), and C-SCAN (Circular SCAN, provides uniform wait times).